MyYanga v4 - Functionality Architecture & Platform Capabilities

The best of African fashion creativity, in one spot.

MyYanga is the portal where lovers of African fashion discover, engage and patronize African creatives - designers, photographers, makeup artists and fabric makers. Version 4 is the pre-production release: monetizable end-to-end, with AI Virtual Wear from a **paid brand pool**, member Settings for identity + reel scheduling, multi-parameter Smart Search, competitions, editorial programming, anonymized analytics with brand Insights, and a full admin console with delegated rights.

Prepared: July 2026 - ZonicMe Limited - Abuja - Lagos - Diaspora Hostile audit: **80 pass / 0 fail - 100/100 - SHIP** (npm run audit:aggressive)

1. Product Vision

> *Discover African creatives. Engage with their work. Patronize them directly. See yourself in their fashion before you buy - and let fashion reels find you on your schedule.*

MyYanga connects three audiences:

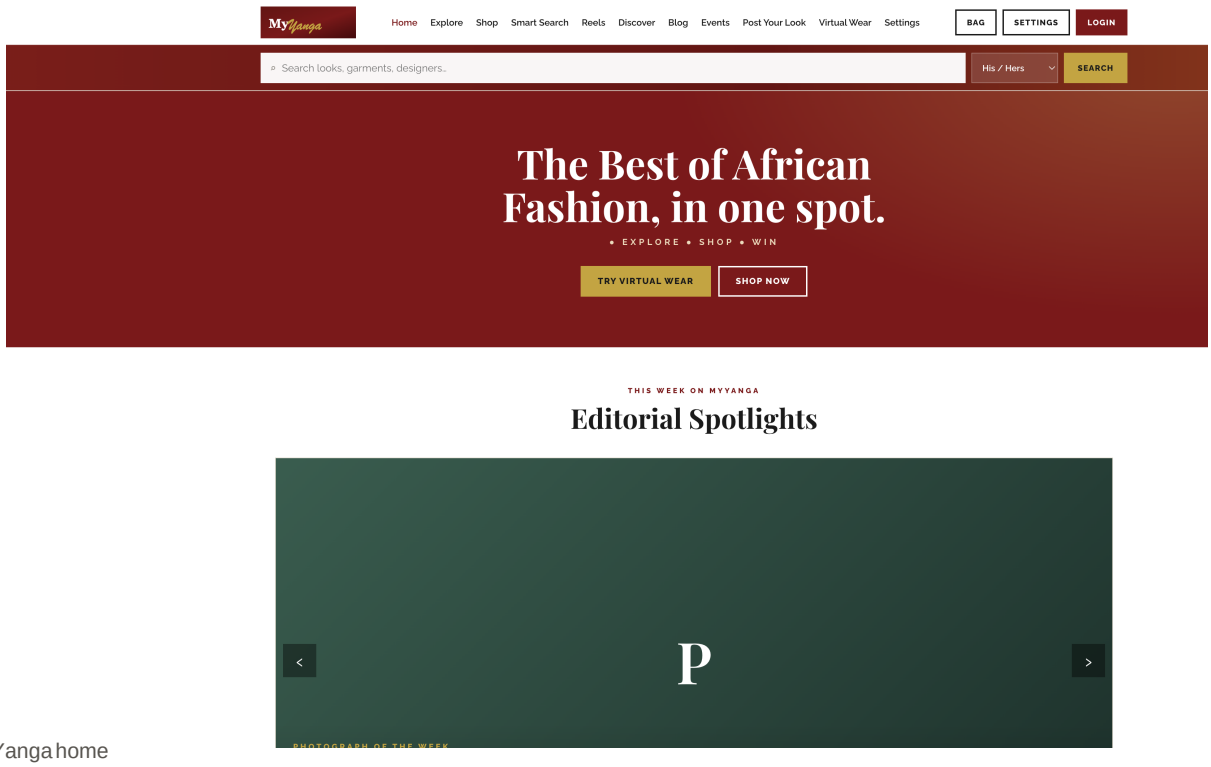
- **Fashion lovers** - browse editorial spotlights, shop looks, try garments virtually from promoted brands, receive curated reels, enter competitions, share looks with friends.
- **African creatives / brands** - pay into promotion to appear on Discover and rotate into Virtual Wear; gain credibility-ranked storefronts and direct patronage (WhatsApp, Instagram, shop).
- **Platform operator (ZonicMe)** - every feature free or paid, priced from Admin; payments across Flutterwave -> Paystack -> Stripe; analytics without selling raw PII.

User advantages (why members stay)

Advantage	How MyYanga delivers it
See yourself before you buy	Virtual Wear: photo -> His/Hers + event theme + outfit type -> randomized look from paid brands; no outfit repeat until the pool cycles
Control when fashion finds you	Settings: channels (inbox / email / push / WhatsApp), cadence, preferred days, local time, timezone, quiet hours
One search, many dimensions	Smart Search: His/Hers + country, event, style, colour, piece, price band - every tick narrows together
Patronize creatives directly	Discover promoted pool with equal WhatsApp / Instagram / Virtual Wear / Shop actions
Privacy-respecting analytics	Site events are anonymized; member profile details are never sold as raw PII in analytics
Sign in the way Africa actually signs in	Google default; WhatsApp OTP secondary (Meta Cloud API, **no BSP**); email tertiary

2. Visual walkthrough

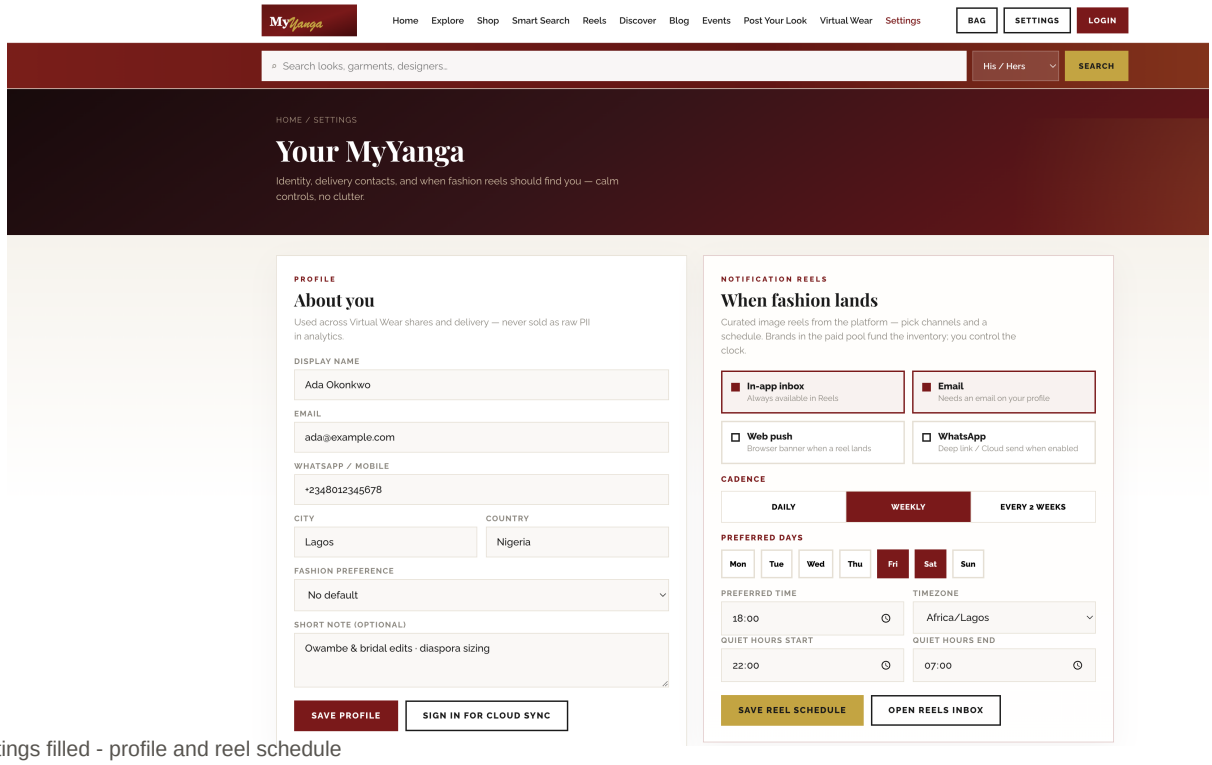
2.1 Home - brand-first entry



MyYanga home

Hero CTA pair: **Try Virtual Wear** and **Shop now**. Sticky header exposes Settings beside Login. Editorial spotlights and category rails follow.

2.2 Settings - identity + notification reel schedule (sleek member hub)



Settings filled - profile and reel schedule

Your MyYanga is a two-panel composition:

Left - Profile / About you

- Display name, email, WhatsApp/mobile, city, country

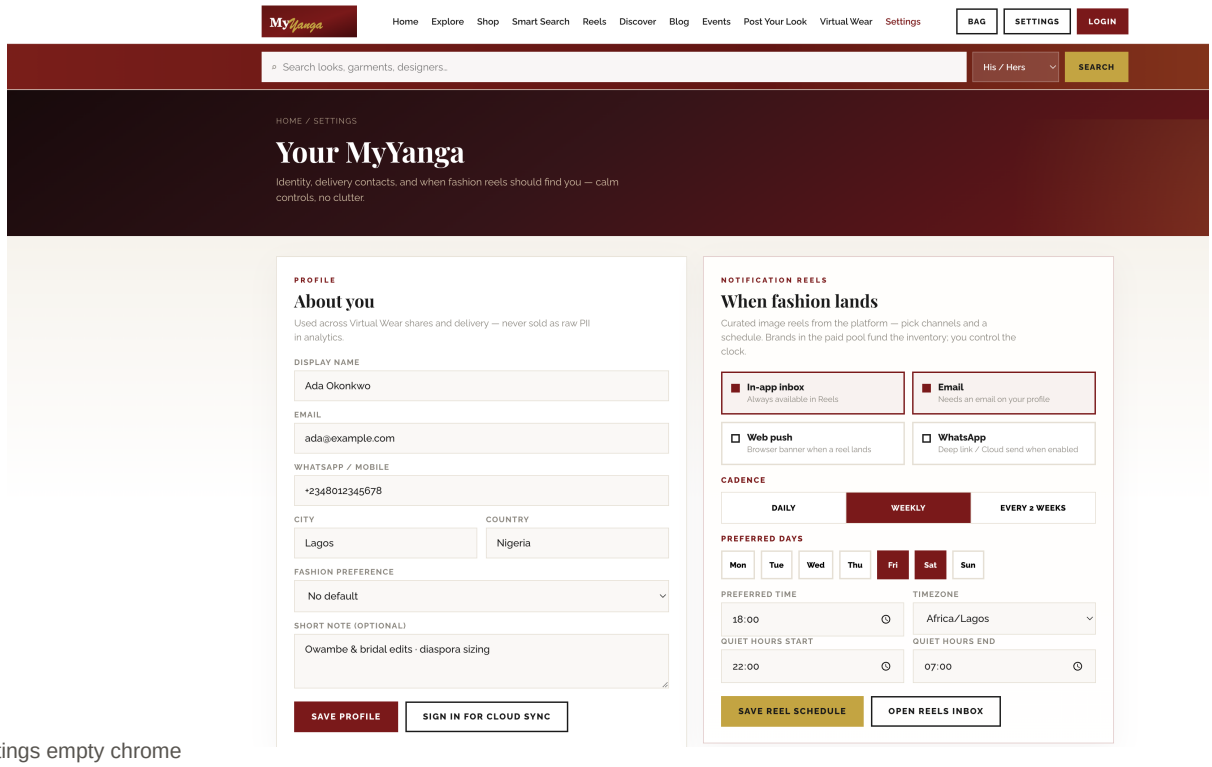
- Fashion preference (Hers / His / none)
- Short style note (occasions, diaspora sizing)
- Save profile - optional Sign in for cloud sync

Right - Notification reels / When fashion lands

- Channel tiles: In-app inbox - Email - Web push - WhatsApp
- Cadence: Daily - Weekly - Every 2 weeks
- Preferred days (Mon-Sun chips)
- Preferred time + timezone (Africa/Lagos default; Accra, Johannesburg, London, New York)
- Quiet hours start/end
- Save reel schedule - Open Reels inbox

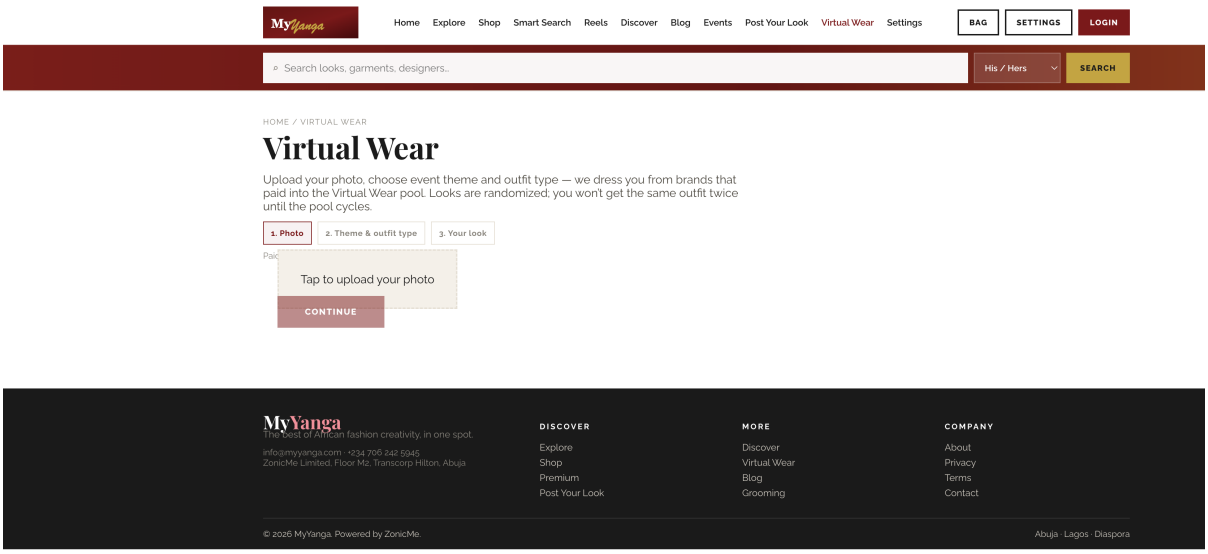
Copy states the business model clearly: *brands in the paid pool fund the inventory; the member controls the clock.*

Empty / first-visit chrome (same layout, placeholders):



Settings empty chrome

2.3 Virtual Wear - paid pool rotator

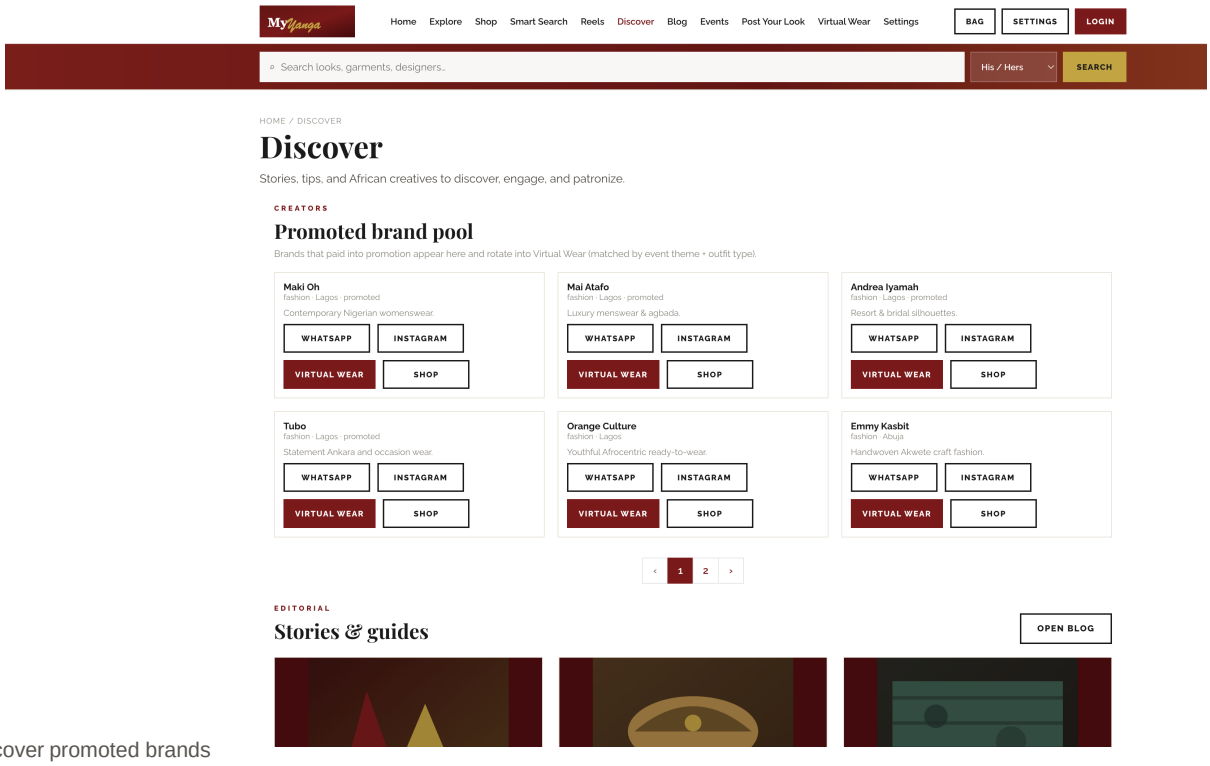


Virtual Wear photo step

Three-step flow: **1. Photo -> 2. Theme & outfit type -> 3. Your look**.

Member copy: upload a photo, choose event theme and outfit type; the app dresses them from brands that **paid into the Virtual Wear pool**. Looks are randomized; the same outfit does not return until the matching pool is exhausted.

2.4 Discover - promoted brand pool

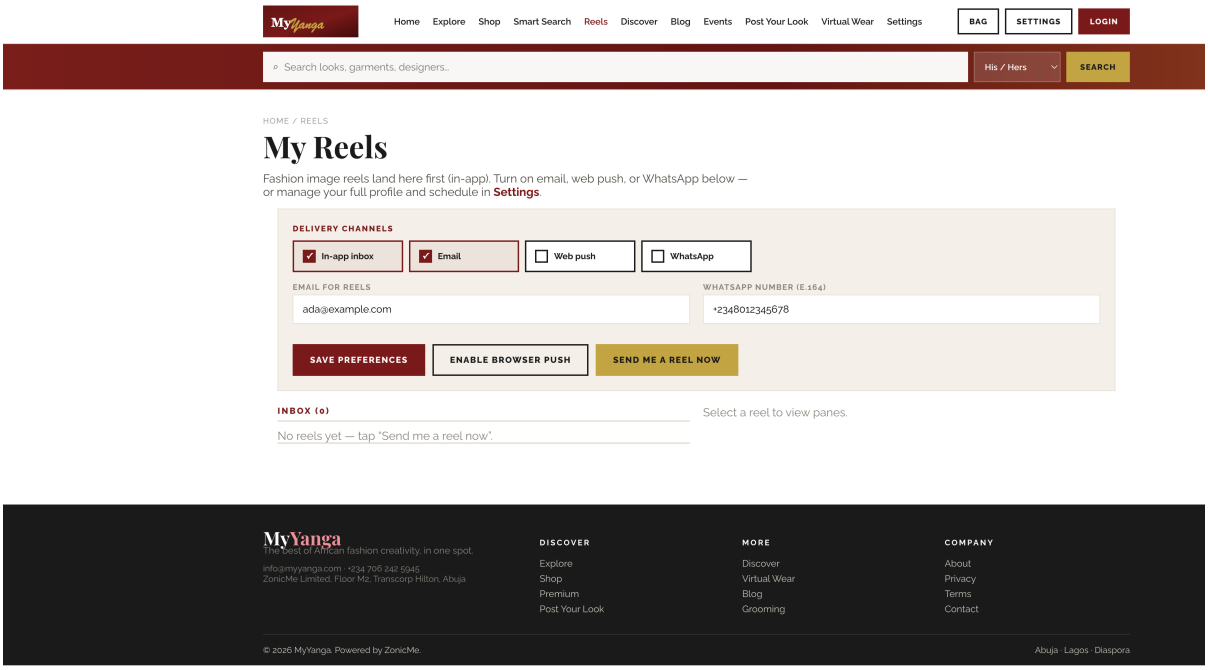


Discover promoted brands

Creators with is_promoted appear here and rotate into Virtual Wear (matched by gender + event theme + outfit

type). Each card exposes equal actions: WhatsApp, Instagram, Virtual Wear, Shop.

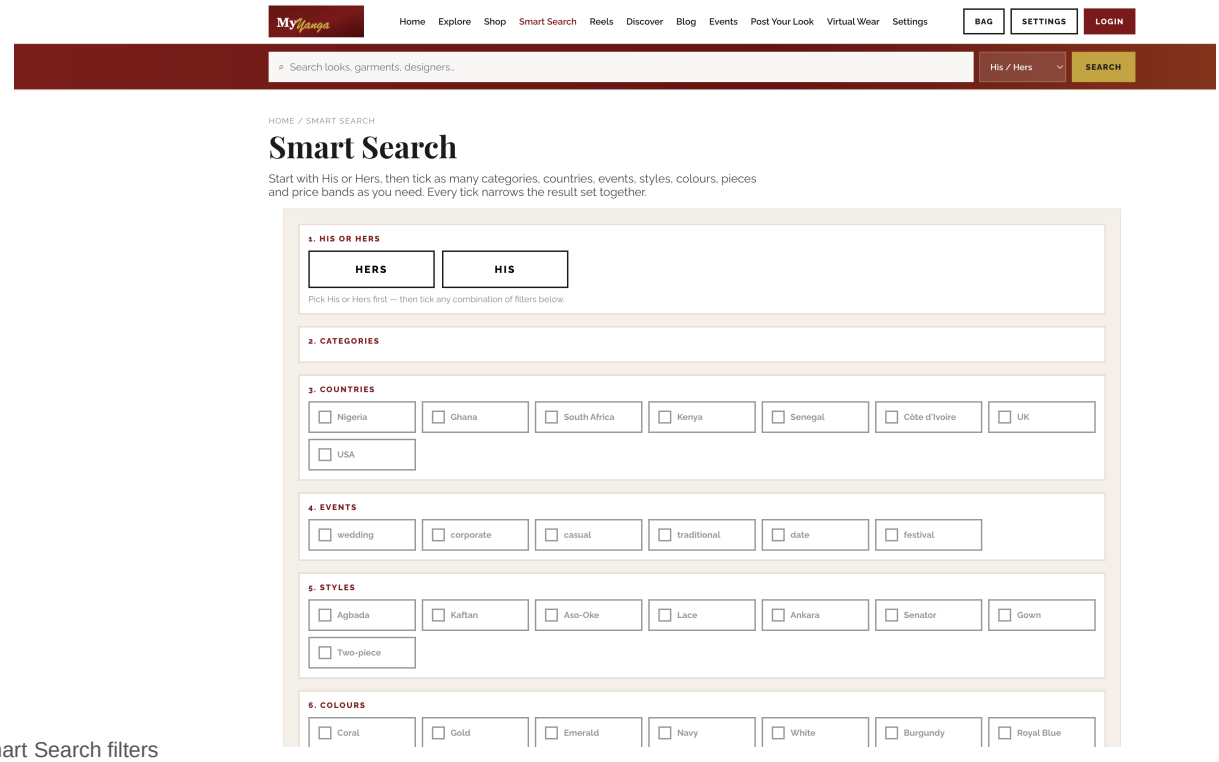
2.5 Reels inbox - delivery surface



My Reels inbox

In-app inbox is the always-on destination. Quick channel toggles live here; full schedule and profile live in **Settings** (linked from the page lead).

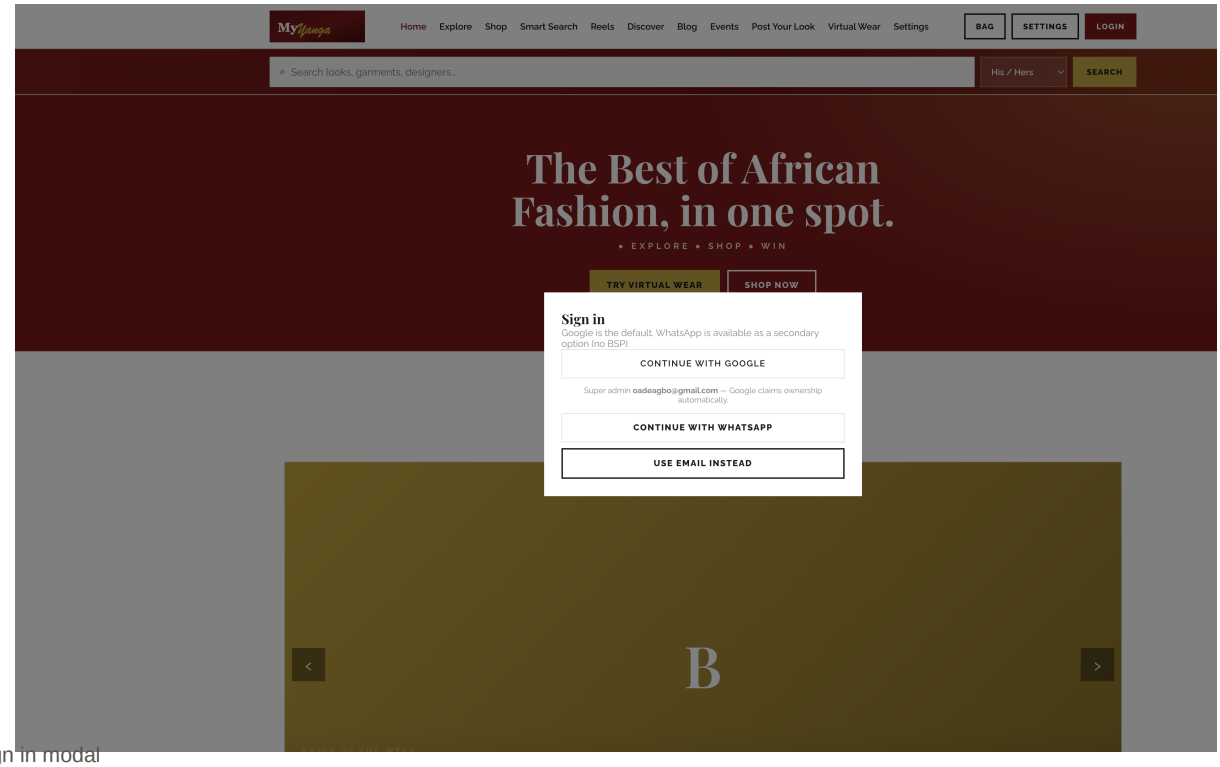
2.6 Smart Search - multi-parameter discovery



Smart Search filters

Start with His or Hers, then tick countries, events, styles, colours, pieces and price bands. Returns garments and assembled look decks (brand, per-item cost, total).

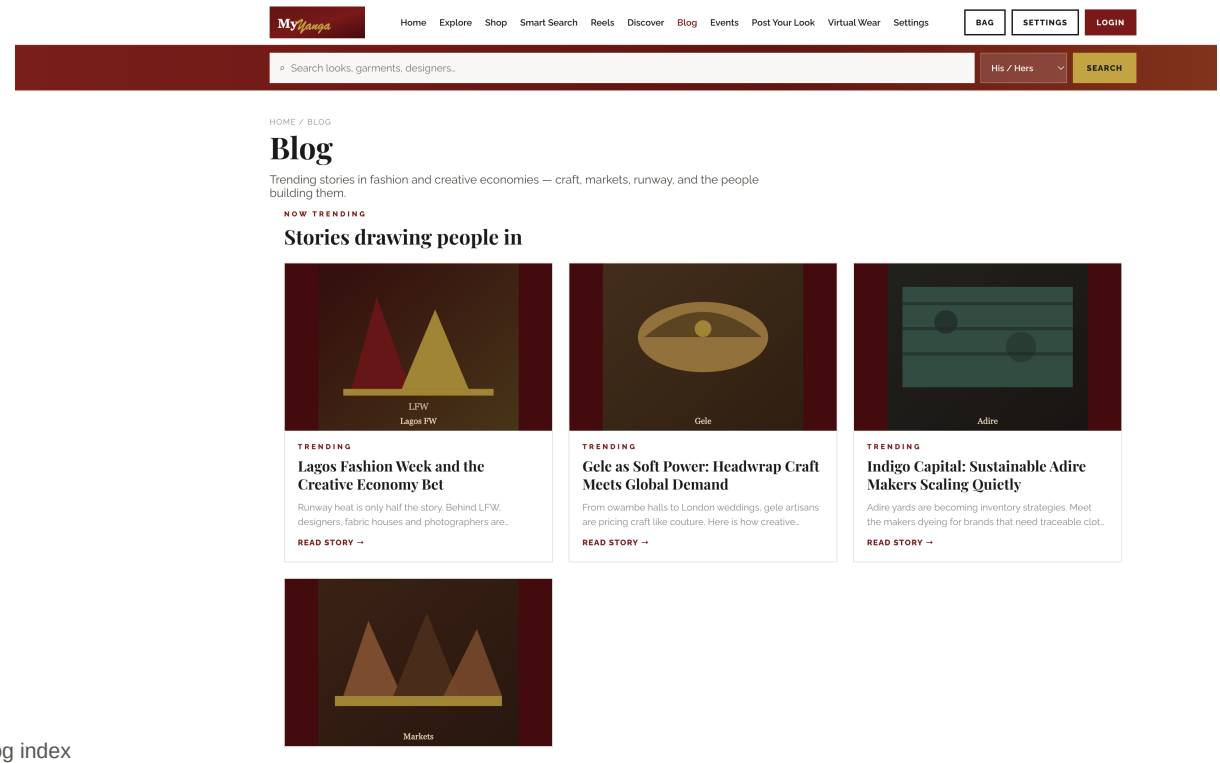
2.7 Sign-in - Google default, WhatsApp secondary



Sign in modal

Continue with Google (primary) - Continue with WhatsApp (no BSP) - Use email instead. Founding Google address claims super admin automatically.

2.8 Blog - acquisition funnel



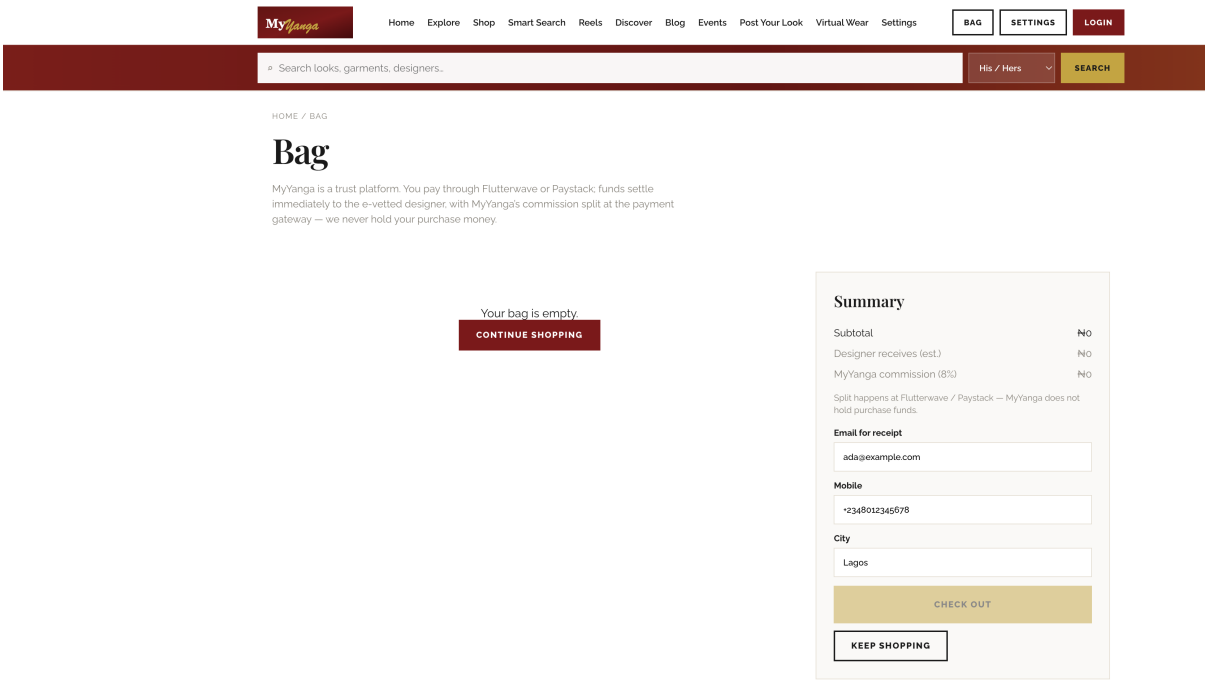
Blog index



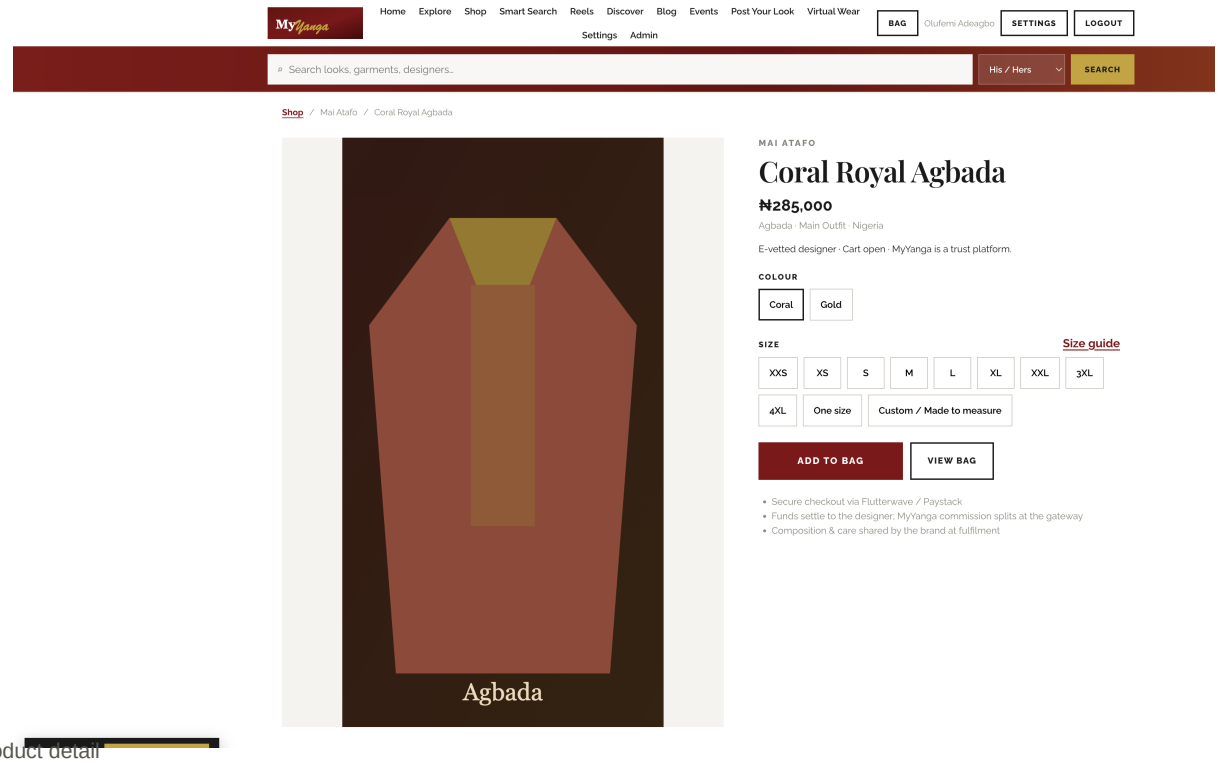
Article

Index teasers -> article with share + guest comments + **Explore MyYanga** CTA.

2.9 Bag & PDP



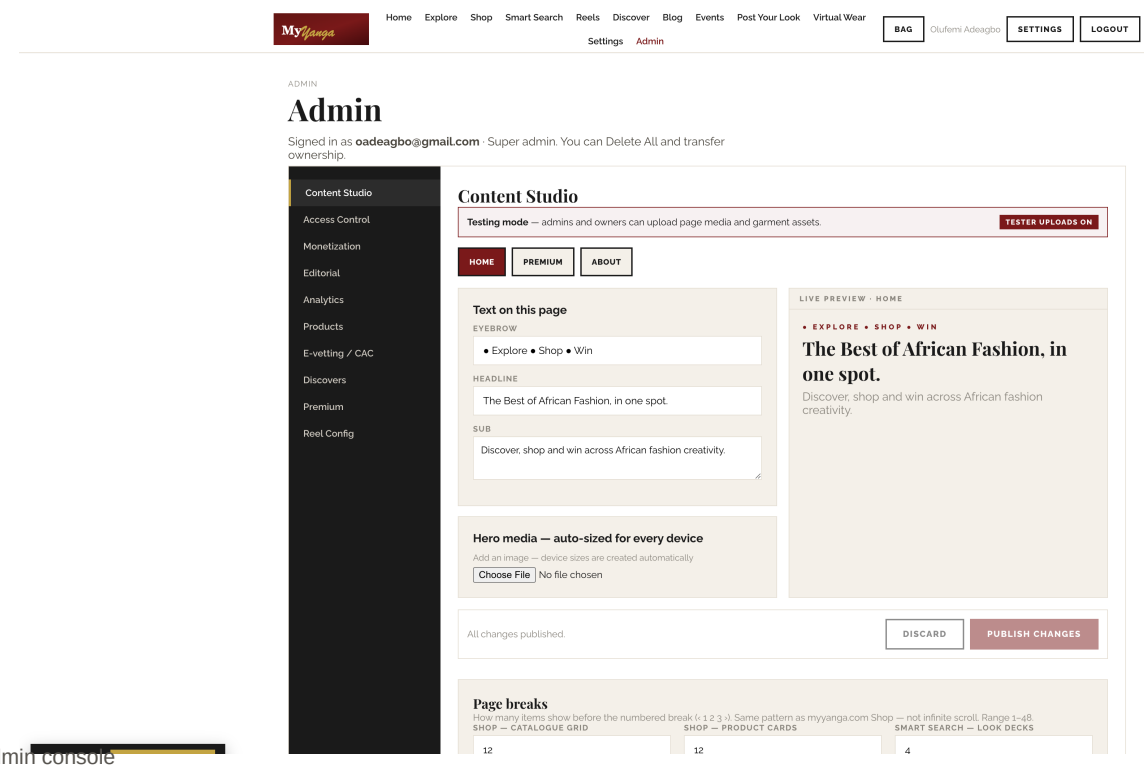
Bag



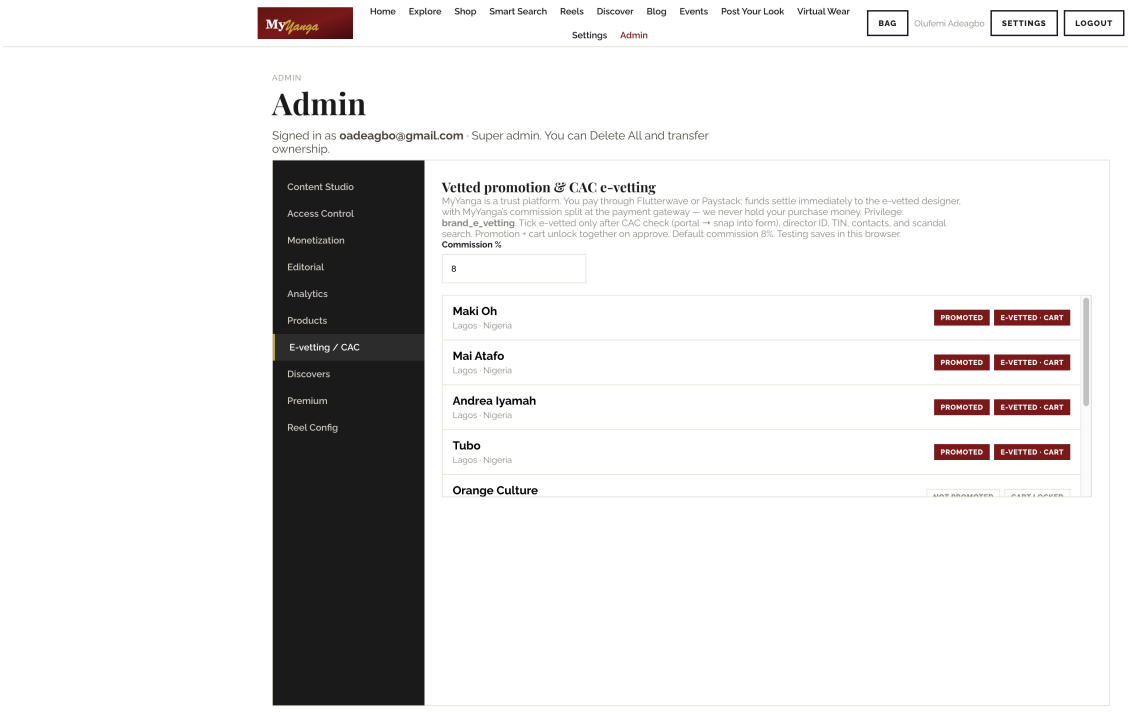
Product detail

Apple-minimal PDP; Bag with designer split preview. Add to Bag only for e-vetted brands.

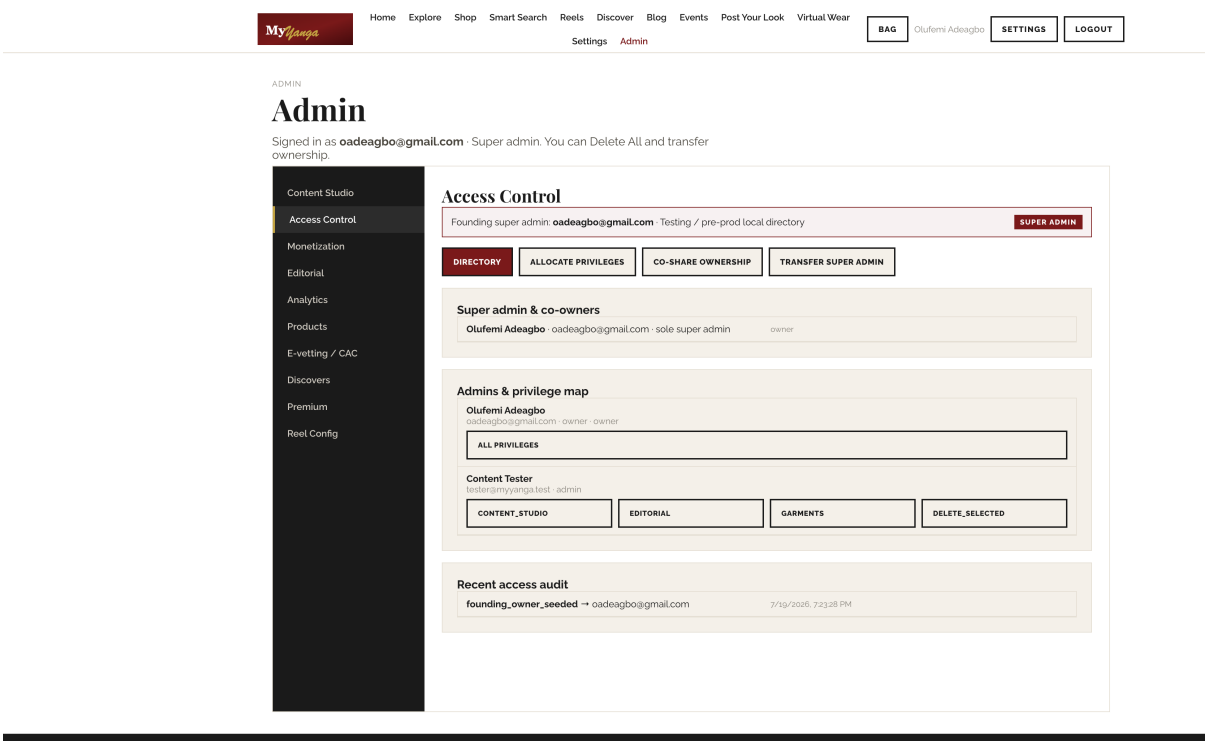
2.10 Admin surfaces (operator)



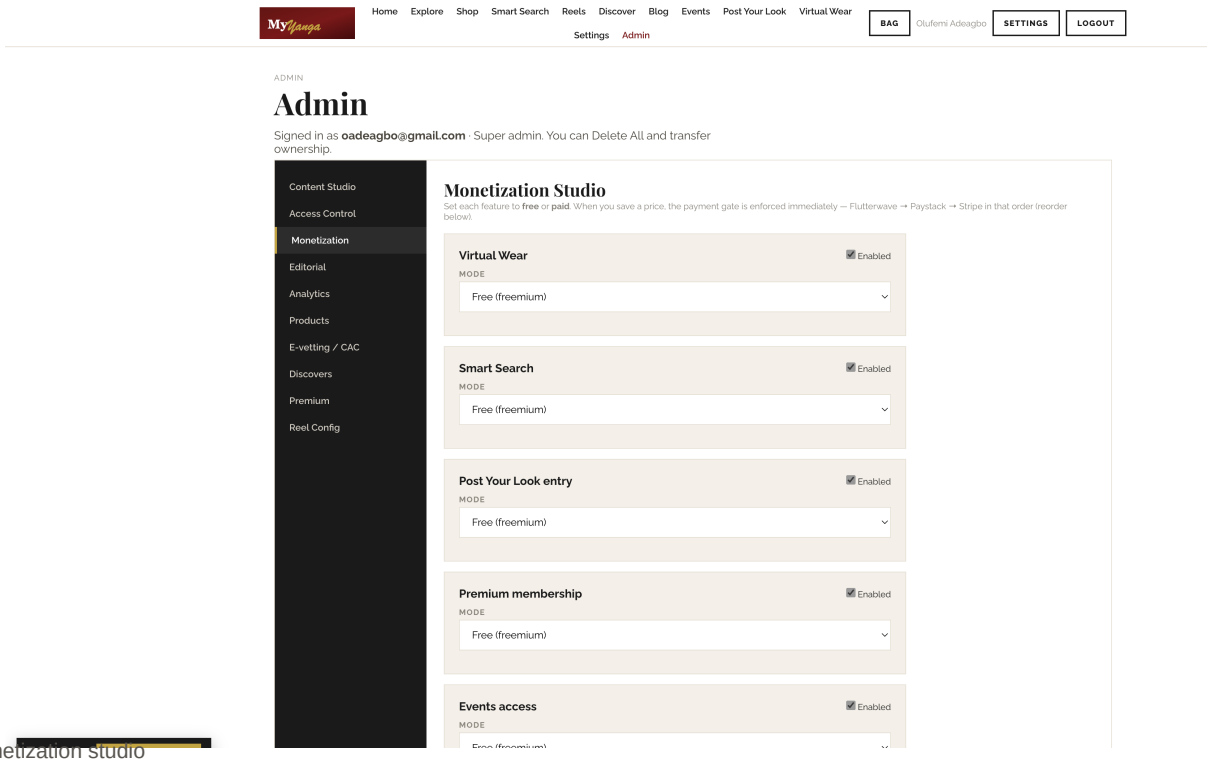
Admin console



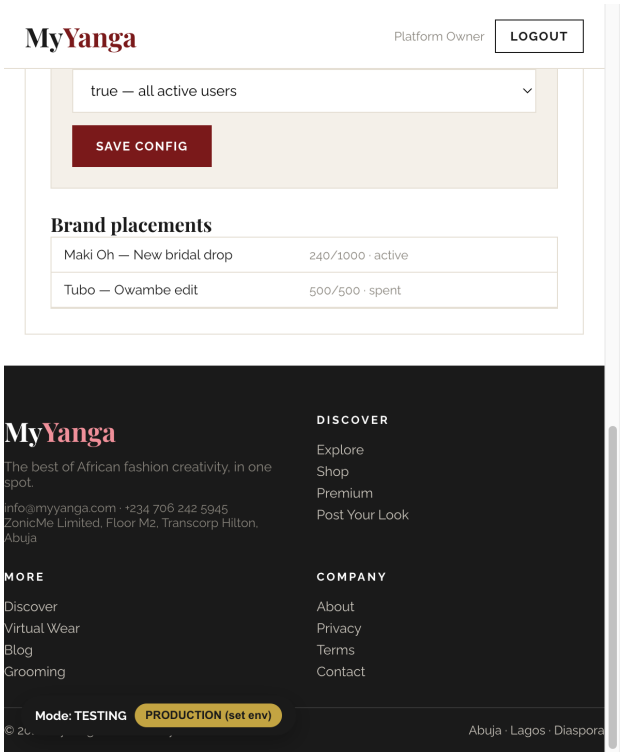
E-vetting / CAC



Access control



Monetization studio



Reel config

3. Feature catalogue

3.1 Discovery & editorial

Feature	What it does
Editorial carousel	Outfit of the Day, Bride/Groom/Couple of the Week, Photograph / Fabric / Makeup / Designer in Focus - admin-published
Blog	Acquisition funnel: nav **Blog**, home + Discover teasers (title, 2-line excerpt, thumbnail) -> article with share + guest comments + **Explore MyYanga** CTA back to home
Discover	Stories & guides (wired to Blog articles) + **promoted brand pool** (is_promoted)
Explore & Shop	Category browse and shoppable product grid
Events	Fashion events with venue, city, schedule; optionally pay-gated

3.2 Virtual Wear (AI try-on) - paid brand pool

- Member uploads a photo
- Chooses **His/Hers**, **event theme** (wedding, owambe, corporate, traditional, date, casual,) and **outfit type** (Agbada, Gown, Ankara,)
- App picks a **random** main_outfit from brands with **is_promoted** (paid Virtual Wear / Discover promotion), matching gender + event tag + style
- Same outfit is **never repeated** for that wearer until the matching pool is exhausted (then recycles) - local no-repeat store (vwPool.js)
- Optional same-brand accessories; Gemini AI render with on-device canvas fallback

Monetization gate: creator_promotion -> labelled **Virtual Wear + Discover promotion**. Brands pay to be in the pool; members experience fair rotation, not a designer picker.

3.3 Shop - PDP, Bag & trust checkout

MyYanga is a **trust platform**, not the merchant of record holding purchase money.

Piece	Behaviour
Catalog -> PDP	Apple-minimal layout; Selfridges-complete colour + size matrix + size guide
E-vetting gate	**Add to Bag** only when brand is is_e_vetted (Admin -> E-vetting / CAC)
Bag	Sparse summary, qty, designer split preview, email/city
Settlement	Flutterwave multi-subaccount split (or Paystack single-brand); **commission % at gateway**
Trust	Funds settle to the designer immediately; MyYanga commission is gateway-split - we never hold the bag total
Testing	Mock settle records a local order and clears the bag

Feature-gate payments (Virtual Wear unlock, Smart Search,) stay separate from retail checkout.

Brand e-vetting (CAC) - unlocks cart

Privilege: **brand_e_vetting** (Access Control). Admin opens **Admin -> E-vetting / CAC**:

- Open CAC portal / search brand RC
- Paste listing -> **Snap listing into form** (legal name, RC, address, director, TIN)
- Complete contacts, socials, adverse-media / scandals notes
- Upload director **ID shot**
- Tick **CAC verified** + **scandals cleared**
- **Approve e-vet + unlock cart** -> sets is_e_vetted + is_promoted; Flutterwave subaccount required for live settlement

Minimums enforced in UI before approve. Schema: migration 009 (commerce) + 010 (CAC dossier columns).

3.4 Member Settings (profile + reel clock)

Area	Controls	Persistence
Profile	Name, email, phone, city, country, fashion preference, bio	myyanga_member_settings (Testing); cloud profile when signed in
Channels	In-app, email, web push, WhatsApp	myyanga_reel_prefs
Schedule	Frequency, days, time_local, timezone, quiet_start / quiet_end	Nested under reel prefs
Actions	Save profile, save schedule, enable push, open Reels	Analytics: settings_save_profile, REEL_CHANNEL

Business effect: delivery inventory is brand-funded; attention timing is member-owned - higher engagement, lower fatigue, clearer monetization story for paid pool brands.

3.5 Notification reels

Configurable image reels: in-app inbox (live), email (Resend-ready), web push, WhatsApp queue + wa.me / Meta Cloud (no BSP). Admin Reel Config meters brand ad slots. Member schedule in Settings gates *when* delivery is appropriate.

3.6 Look sharing & competitions

- Share / download generated looks; public /look/{code} with guest comments (RLS)
- Post Your Look: votes via vote_pyl; entry optionally gated (pyl_entry)
- Blog articles: /?view=blog index + /?view=article&slug=; Web Share / copy link; guest comments via blog_comments (migration 011) or Testing localStorage

3.7 Monetization - freemium per feature

Gate	Feature
virtual_wear	Virtual try-on
smart_search	SmartSearch AI
pyl_entry	Competition entry
premium	Premium membership
events	Event access
creator_promotion	Paid placement in Discover + VW pool
retail_checkout	Marketplace bag (gateway split - not an entitlement gate)

Providers cascade: **Flutterwave -> Paystack -> Stripe**. Feature entitlements verified on return and webhooks. Retail uses the same providers with **subaccount split**.

3.8 Analytics & Insights

- Anonymized autocapture + fashion-granular events (analytics.js, migration 008)
- Admin Analytics + Insights studio; brand-scoped keys via creators.id
- Orbit export path for monetized brand reporting

See also: docs/FASHION_ANALYTICS.md.

4. Administration & governance

Studio	Purpose
Content Studio	Page copy/media, garment uploads in Testing
Access Control	Directory, privileges, co-owners, transfer, audit
Monetization	Per-feature free/paid + provider order
Editorial	Carousel programming
Analytics / Insights	Site KPIs + fashion rollups + AI assistant
Catalog	Products, discovers, premium
E-vetting / CAC	Privilege brand_e_vetting: CAC snap, KYC dossier, promote, unlock cart + subaccounts
Reel Config	Reel structure + brand placements

Auth: Google default - WhatsApp OTP secondary - email tertiary. Founding owner: oadeagbo@gmail.com. Rights mutations are SECURITY DEFINER RPCs with audit - UI is convenience, not the boundary.

5. Testing vs Production

	Testing (default)	Production
Data	Seeded mocks + localStorage uploads	Supabase Postgres + RLS
Auth	Google / WhatsApp / email; founding = owner; testers get Admin Studio	DB roles; WhatsApp via auth-whatsapp edge
Pay gates	All free	Live gates + payments
Retail cart	Mock Flutterwave split settle	Live subaccount split via payments edge
Virtual Wear	Canvas compositing	Gemini + canvas fallback
Settings / reels	Local prefs + inbox	Same UX; cloud sync when signed in

6. Technical architecture

6.1 Stack

Layer	Technology
Frontend	React 18 + Vite 5, bespoke design system (Playfair / Raleway, maroon-gold-cream)
Backend	Supabase: Postgres + RLS, Auth, Storage, Edge Functions (Deno)
Payments	Flutterwave, Paystack, Stripe
AI	Gemini 2.5 Flash Image (try-on), Anthropic Claude (stylist / insights)
Hosting	Static dist/ + Supabase data plane

6.2 Client seams (selected)

- src/myyanga-app.jsx - shell, routing (view= + /look/:code + blog/article slugs), design system, admin
- src/pages/BlogPage.jsx + src/lib/blogPosts.js + src/lib/blogComments.js - editorial acquisition blog
- src/pages/SettingsPage.jsx - sleek profile + reel schedule
- src/pages/VirtualWearPage.jsx + src/lib/vwPool.js - paid pool rotator + no-repeat
- src/lib/memberSettings.js, reelStore.js, reelEngine.js - prefs + delivery
- src/lib/analytics.js - anonymized events
- src/lib/dataBridge.js - testing <-> production
- src/lib/accessControl.js, monetization.js, virtualWear.js, smartSearch.js

6.3 Database & edge

Migrations 001-011 (core -> access control -> analytics -> commerce -> e-vetting -> blog). Edge: payments, smart-search, virtual-tryon, ai-chat, send-reel, auth-whatsapp - JWT + server-side gate checks.

6.4 Security posture

- Production roles only in DB; no client role map
- Profiles RLS + safe public view; self-update cannot elevate ownership
- Edge secrets never in client; no demo passwords in production UI

- Typed confirmation for delete-all and super-admin transfer

6.5 Quality gates

- npm run verify:production
- npm run audit:aggressive - **80/80 - SHIP** (covers VW pool, Settings, analytics, WhatsApp auth, payments, blog funnel, footer legal pages, business/features docs, build)
- npm run package:staging
- npm run docs:pdf - regenerates this document as PDF with embedded screenshots

6.6 Gemini cost guidance (pass-through to brands)

Model gemini-2.5-flash-image **$0.039/image** + input **NGN 54-85/render** at ~NGN 1,380/USD. Recommended pool pass-back: on the order of **NGN 125,000 per 1,000 successful AI renders**. Canvas fallback is free when AI is unavailable (not a quality substitute for Gemini).

7. Deployment runbook (summary)

- supabase db push (migrations through analytics)
- Secrets: payment providers, GOOGLE_API_KEY, ANTHROPIC_API_KEY, PUBLIC_APP_URL, optional RESEND_API_KEY, WhatsApp Cloud token + phone number id - see docs/AWS_SECRETS_RUNBOOK.md
- npm run sb:deploy
- Enable Google Auth; deploy auth-whatsapp for secondary OTP
- npm run build -> deploy dist/ with SPA fallback for /look/*
- First Google login as founding email -> configure monetization, editorial, promotion seats

Footer Company links (About / Privacy / Terms / Contact) and Grooming resolve to in-app legal placeholders (LegalPage.jsx) - never 404.

8. Known v4 limits & roadmap

- Shop catalog browse + **PDP / Bag / Flutterwave split checkout** for e-vetted brands (trust model - MyYanga does not hold purchase funds; commission splits at the gateway)
- **Blog** acquisition funnel shipped (seeded fashion / creative-economy stories, share, comments, Explore MyYanga); production CMS admin for blog_posts still thin vs editorial carousel studio
- Seeded garment art is placeholder until real photography is loaded via Content Studio
- Webhook signature verification should be deepened before high volume
- Reel email / push / WhatsApp Cloud need production secrets; dedicated SIM alone cannot automate WhatsApp

9. Screenshot index

File	Section
assets/myyanga-home.png	Home hero
assets/myyanga-settings-filled.png	Settings profile + schedule (filled)
assets/myyanga-settings.png	Settings chrome
assets/myyanga-virtualwear.png	VW photo step / paid pool copy
assets/myyanga-discover.png	Promoted brand pool
assets/myyanga-reels.png	Reels inbox + channels

assets/myyanga-smartsearch-results.png	Smart Search
assets/myyanga-blog.png	Blog index
assets/myyanga-article.png	Blog article
assets/myyanga-bag.png	Bag
assets/myyanga-pdp.png	Product detail
assets/myyanga-login.png	Auth modal
assets/myyanga-admin.png	Admin
assets/myyanga-evetting.png	E-vetting / CAC
assets/myyanga-accesscontrol.png	Access Control
assets/myyanga-monetization.png	Monetization
assets/myyanga-reelconfig2.png	Reel Config

Capture refresh: npm run build && node scripts/capture-architecture-shots.mjs && node scripts/build-architecture-html.mjs. Secrets: docs/AWS_SECRETS_RUNBOOK.md. Business: docs/MYYANGA_BUSINESS_PLAN.md. Tech matrix: docs/FEATURES_AND_IMPLEMENTATION.md.

Prepared for ZonicMe Limited - MyYanga v4 functionality architecture. Confidential.